Detecting Political Manipulation as a Defensive Practice: A Transparent Baseline for Explainable Propaganda Detection

A Working Paper in Computational Political Anthropology

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Politomorphism Engine — Defensive Political Anthropology Layer

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Code & data: github.com/politomorphism/politomorphism-defense

Abstract

This paper argues that the automated detection of political manipulation should be framed as a defensive practice rather than as the inverse of a generative one, and it establishes a transparent, reproducible baseline for that practice. Using the English partition of the PropXplain corpus — a multilingual, explanation-annotated dataset for propaganda detection — I train a conventional TF-IDF and logistic-regression classifier to separate propagandistic from non-propagandistic news paragraphs. On a held-out test set of 922 paragraphs, the baseline attains an F1 of 0.506 on the propagandistic class (precision 0.465, recall 0.556), against a class distribution in which non-propagandistic paragraphs account for roughly 73% of the data. I deliberately foreground precision, recall, and F1 on the minority (manipulation) class rather than overall accuracy, which is shown to be misleading under imbalance. An error analysis indicates that the lexical model captures manipulation that is carried by overt vocabulary but systematically misses manipulation carried by presupposition, insinuation, and rhetorical equivalence — a finding that is substantive rather than incidental. I interpret this gap through the lens of Computational Political Anthropology and position the detector as the audit layer of the Politomorphism Engine: an instrument that converts a qualitative theory of manipulation into a measurable quantity. The paper closes by specifying the contextual, transformer-based successor that the baseline motivates, and by situating explainability as the bridge between anthropological interpretation and computational measurement.

Keywords: propaganda detection; computational political anthropology; disinformation; explainable NLP; text classification; defensive auditing; Politomorphism

1. Introduction

Political manipulation is, before it is anything computational, an anthropological phenomenon. It is a practice by which a speaker reorganizes the perceptual and moral field of an audience — directing attention, assigning blame, manufacturing threat, foreclosing alternatives — so that a contingent interpretation comes to feel like a natural fact. The study of how collectivities are shaped through such practices has long belonged to political anthropology. What is new is that the practice now operates at machine scale, through automated text generation, coordinated amplification, and models that can produce persuasive framing on demand. The question this paper takes up is methodological: if manipulation has become computational in its production, what does a computational defense look like, and how should it be built so that it remains faithful to the anthropological object it claims to measure?

A tempting but mistaken answer is to build a generator of manipulation and label it defensive. The temptation is understandable — one cannot catch what one does not understand — but the artifact produced by training a model to generate persuasive manipulation is, at the level of its learned parameters, the same capability whether the stated intent is offense or defense. A model optimized to produce convincing propaganda is a propaganda engine regardless of the sign placed in front of it. Defensive work must therefore be architected differently: not around generation, but around detection, attribution, measurement, and inoculation. The defender is not a reformed attacker but something closer to an epidemiologist of discourse, whose task is to make the mechanism of harm visible and quantifiable without reproducing the harm.

This paper contributes a deliberately modest but honestly reported first instrument toward that end: a transparent baseline classifier that separates propagandistic from non-propagandistic news text. The baseline is not presented as a solved problem. It is presented as a floor — a reproducible reference point whose very limitations motivate the contextual successor it is designed to precede. The contribution is threefold. First, I frame manipulation detection as a defensive practice and specify what that framing excludes. Second, I report a fully reproducible baseline on a public, explanation-annotated corpus, with metrics chosen to be truthful under class imbalance rather than flattering. Third, I conduct an error analysis whose central finding — that lexical models capture overt but not structural manipulation — is itself a theoretical result about the nature of political manipulation, and I situate that result within the Politomorphism Engine as the foundation of a defensive audit layer.

2. Framing: Detection as Defense

The distinction between a generative and a detective stance is not rhetorical housekeeping; it determines what is built. A generative model of manipulation learns to produce text that moves an audience. A detective model learns to recognize, in text it did not produce, the signatures of such movement. The two may consume overlapping knowledge, but they yield different artifacts and carry different risks. The defensive commitment adopted here is that the instrument must at no point be a generator of the harmful capability it studies. In practice this means the model is trained only to classify and explain existing text, never to synthesize manipulative content, and that the data it learns from are annotations of naturally occurring journalism rather than engineered attack material.

This framing also clarifies which metric matters. If the defensive goal is to ensure that manipulation does not pass unseen through an audit, then the quantity of interest is how much real manipulation the instrument recovers and at what cost in false alarms — that is, recall and precision on the manipulation class — not aggregate accuracy. As the next sections show, accuracy is actively misleading in this setting, and an instrument tuned to maximize it would be defensively worthless.

3. Corpus

The experiments use the English configuration of PropXplain, a publicly available, multilingual corpus for explainable propaganda detection released by the Qatar Computing Research Institute. PropXplain is distinctive among propaganda resources in that each paragraph carries not only a label — propagandistic or non-propagandistic — but also a natural-language explanation articulating why the passage is or is not manipulative. This explanation layer is central to the present programme: it is the point at which a classifier's quantitative verdict can be reconnected to the qualitative, interpretive reasoning that characterizes anthropological analysis, and it is what distinguishes an opaque detector from an auditable one.

Each example consists of a sample identifier, an input paragraph drawn from news articles, a binary label, a free-text explanation, and several explanation-quality fields (informativeness, clarity, plausibility, faithfulness) that are populated only for certain partitions. For the baseline reported here I

use only the input text and the binary label; the explanation and quality fields are reserved for the explainable successor model described in Section 7. After removing rows with missing text or label, the English training partition comprises 4,472 paragraphs and the held-out test partition 922 paragraphs.

The class distribution is imbalanced in a way that is both typical of the phenomenon and consequential for evaluation. In the test partition, non-propagandistic paragraphs account for 672 examples (72.9%) and propagandistic paragraphs for 250 (27.1%). Manipulation is, in a representative sample of journalism, the minority condition. Any honest evaluation must be built around that fact.

4. Method

The baseline is deliberately conventional and fully transparent. Each paragraph is represented using term frequency–inverse document frequency (TF-IDF) features over unigrams and bigrams, with lowercasing, a minimum document frequency of two, and sublinear term-frequency scaling. These features are passed to a logistic-regression classifier. Two design choices deserve comment. The feature representation is lexical by construction: it encodes which words and short word-pairs are distinctive of each class, and nothing about discourse structure, presupposition, or context beyond the bigram window. This is a limitation, and it is an intended one — the baseline exists precisely to establish how far purely lexical evidence can carry the task, so that a later contextual model can be credited only with what it adds.

The second choice concerns class imbalance. The classifier is fit with balanced class weights, so that errors on the minority (propagandistic) class are penalized in inverse proportion to its frequency. Without this correction, the optimizer is rewarded for neglecting the minority class, since a model that predicts the majority label everywhere already achieves high accuracy. Balancing shifts the objective toward recovering the manipulation class, which is the defensively relevant behavior. The model is trained on the 4,472-paragraph training partition and evaluated once on the untouched 922-paragraph test partition; no test data inform training or feature selection, so the reported figures are an out-of-sample estimate.

5. Results

Table 1 reports the full per-class classification results on the held-out test partition. The baseline attains an F1 of 0.506 on the propagandistic class, with precision 0.465 and recall 0.556. On the non-propagandistic class it reaches an F1 of 0.791. Overall accuracy is 0.706.

Class	Precision	Recall	F1	Support
non-propagandistic	0.822	0.762	0.791	672
propagandistic	0.465	0.556	0.506	250
Accuracy			0.706	922
Macro avg.	0.643	0.659	0.649	922
Weighted avg.	0.725	0.706	0.714	922

Table 1. Per-class performance of the TF-IDF + logistic-regression baseline on the PropXplain English test partition (n = 922). The propagandistic row is the defensively relevant one.

The accuracy figure of 0.706 is, taken alone, worse than useless as a summary of defensive quality — and seeing why is the point. A trivial classifier that labels every paragraph non-propagandistic would score 0.729 accuracy on this test set, higher than the trained model, while detecting no manipulation whatsoever. Accuracy rewards exactly the behavior a defensive instrument must avoid. This is why the propagandistic-class F1 of 0.506, unimpressive though it looks beside the inflated accuracy, is the

honest headline figure: it reflects the model's actual ability to recover manipulation, balanced against its tendency to raise false alarms.

The confusion matrix in Table 2 makes the defensive stakes concrete. Of the 250 genuinely propagandistic paragraphs in the test set, the model correctly flagged 139 and missed 111. Of everything it flagged as propaganda, 139 of 299 were correct, the remaining 160 being non-propagandistic paragraphs raised as false alarms. For an auditor, the 111 missed cases are the most important cell in the table: these are the instances of manipulation that would pass undetected through an automated screen built on this model.

	Pred. non-prop.	Pred. prop.
True non-prop.	512	160
True prop.	111	139

Table 2. Confusion matrix on the test partition. Rows are true classes, columns are predictions. The 111 false negatives (bottom-left) are the manipulation an audit would miss.

These numbers are consistent with what a lexical baseline should achieve on an imbalanced, discourse-level manipulation task, and they are reported without adjustment. The value of the baseline is not in the magnitude of the score but in its honesty and reproducibility: it fixes the floor against which a more capable model must prove its worth.

6. Error Analysis

The most instructive result is not a number but a misclassification. Among the test paragraphs the model wrongly labelled non-propagandistic was a passage that drew an equivalence between an official account of an event and a prior contested narrative, through insinuation rather than through any overtly loaded vocabulary. The paragraph is annotated as propagandistic in the corpus; the model saw nothing propagandistic in it. The reason is diagnostic. The manipulation in such a passage does not reside in individual charged words that TF-IDF could key on. It resides in structure — in the presupposition smuggled into a subordinate clause, in the rhetorical equivalence drawn between two unlike things, in what the sentence asks the reader to accept without stating it. A representation built from word and bigram frequencies is, in principle, blind to this.

This is not a defect to be apologized for; it is the paper's central empirical claim stated in miniature. Fine political manipulation is frequently a contextual and structural phenomenon rather than a lexical one. It follows that lexical models establish a genuine but low ceiling, and that meaningful defensive gains require models capable of representing context — the dependency between a clause and its presupposition, the relation between a framing and the thing it frames. The baseline's failures therefore do not merely indicate that a bigger model is needed; they specify what the bigger model must be able to do, and they supply the qualitative evidence that a defensive programme in political anthropology should be organized around structural rather than lexical signatures of manipulation.

7. Positioning within the Politomorphism Engine

The Politomorphism Engine is a framework in Computational Political Anthropology concerned with how political collectivities form, cohere, and fracture — including the dynamics of digital tribalism, elite formation, and disinformation. Much of that framework is descriptive and qualitative: it characterizes the mechanisms by which manipulation operates. The detector introduced here is proposed as the framework's defensive audit layer, and its role is to convert those qualitative characterizations into a measurable quantity. Given a corpus of real political text, an instrument of this kind yields a quantitative map of manipulation density — across actors, channels, and time — turning a

theory of manipulation into a measurement of it. That move, from conceptual to computational, is the one a framework must make to earn the adjective 'computational' rather than merely claiming it.

The programme extends along two coupled lines that share a single technical artifact. The first is detection: the successor to this baseline is a contextual, transformer-based classifier that can represent the structural manipulation the lexical model misses, fine-tuned on the same corpus and credited precisely with the margin by which it exceeds the floor reported here. The explanation layer of PropXplain becomes central at this stage, allowing the detector to report not only that a passage is manipulative but, in the corpus's own terms, why — restoring the interpretive dimension that anthropological analysis requires and that a bare label discards. The second line is evaluation: the same detector, turned on the outputs of generative models, becomes a metric for whether an AI system produces manipulative political framing, which connects the defensive audit layer directly to the evaluation of AI systems on political and media content. A single instrument thus feeds two publications and consolidates the framework as a system rather than a collection of essays.

8. Limitations

Several limitations bound the claims made here. The baseline is lexical and therefore structurally blind, as the error analysis shows; its scores should be read as a floor, not an estimate of achievable performance. The corpus is drawn from news journalism and annotated under a particular guideline set, so both the label distribution and the operative definition of 'propagandistic' are specific to that source and should not be assumed to transfer without revalidation to social-media text, other languages, or other political contexts. Binary labelling collapses a rich taxonomy of techniques into a single axis; the finer question of which technique is present is deferred. Finally, the detector measures textual signatures of manipulation and makes no claim about intent, coordination, or effect, each of which requires evidence beyond the text itself.

9. Conclusion

A defensive computational practice for political manipulation should be built around detection, measurement, and explanation rather than around a generator wearing a defensive label. This paper has established a transparent, reproducible baseline for that practice, reported it honestly under class imbalance, and shown through error analysis that its principal failure — blindness to structural, non-lexical manipulation — is a substantive finding about the phenomenon rather than a mere shortfall in performance. That finding specifies the contextual successor the baseline is designed to motivate and anchors a defensive audit layer within the Politomorphism Engine, where the instrument converts an anthropological theory of manipulation into a measurable quantity and couples the detection of manipulation to the evaluation of the systems that increasingly produce it.

Reproducibility

All results in this paper are produced by a short, self-contained pipeline: data acquisition and inspection, label inspection, and a single training-and-evaluation script using TF-IDF features and logistic regression as implemented in scikit-learn. The reported figures correspond to one evaluation on the held-out test partition and are reproducible from the public PropXplain English configuration. The complete code, pinned dependencies, and this paper are maintained in the public repository at github.com/politomorphism/politomorphism-defense.

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